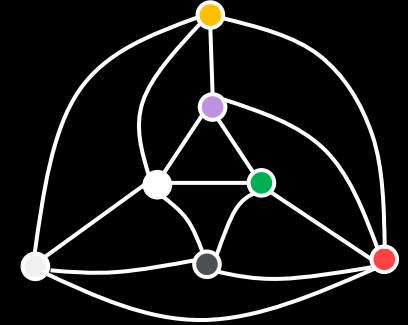


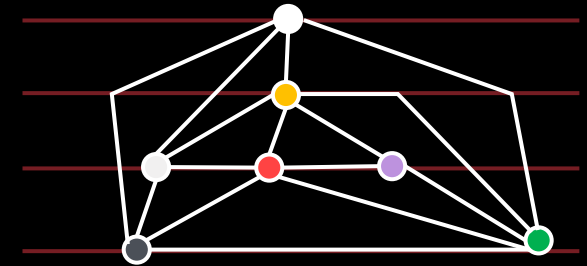
Layered Polyline Drawings of Planar Graphs

Debajyoti Mondal

Department of Computer Science



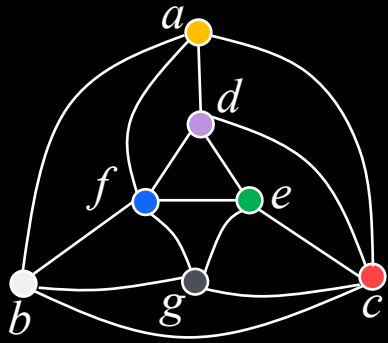
A planar graph G



A layered polyline drawing of G

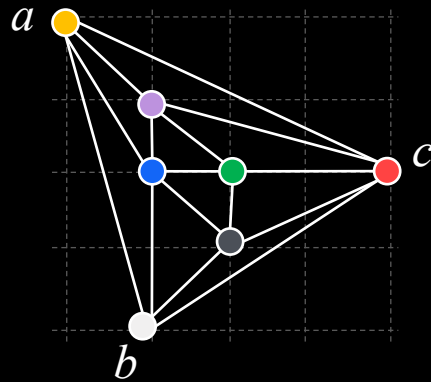
Drawing Models

Input



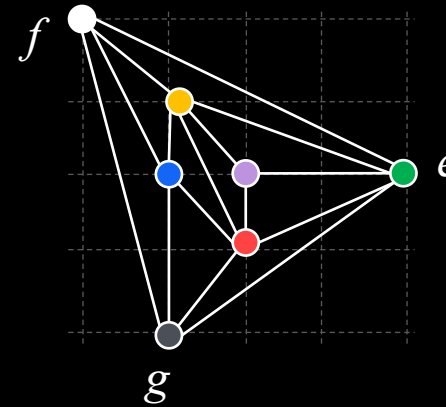
A planar graph G

Straight-line Drawing
(fixed embedding)



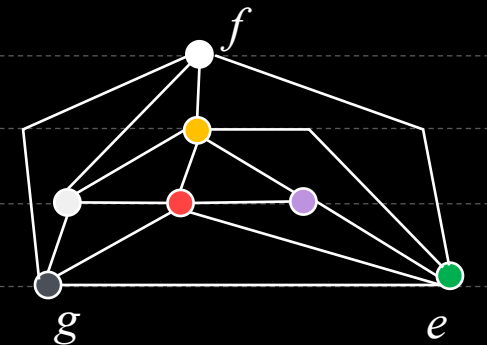
A grid drawing on 5 x 5 grid

Straight-line Drawing
(variable embedding)

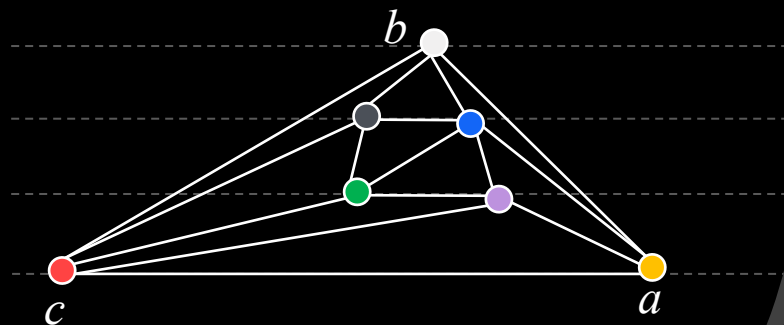


A grid drawing on 5 x 5 grid

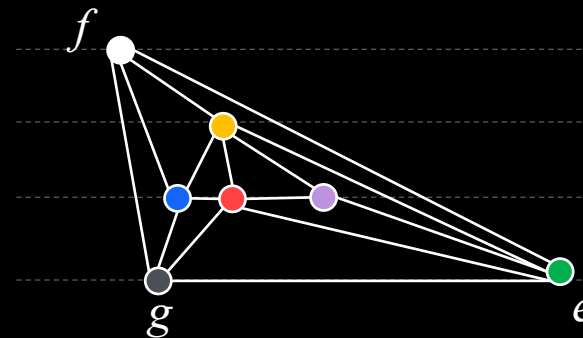
Layered Polyline Drawing
(variable embedding)



This Presentation



A layered drawing on 4 layers



A layered drawing on 4 layers

Known Results

Fixed Embedding

Upper Bounds

<u>Area</u>	<u>Layer</u>		
$2n^2 + O(1)$	$n - 2$	(T)	[de Fraysseix et al. 1990]
$n^2 + O(1)$	$n - 2$	(T)	[Schnyder 1990]
$1.78n^2 + O(1)$	$0.66n$	(T)	[Chrobak and Nakano 1998]
$0.88n^2 + O(1)$	$0.66n$	(T)	[Brandenburg 2008]
$0.44n^2 + O(1)$	$0.66n$	(T)	(polyline) [Bonichon et al. 2003]

Lower Bounds

<u>Area</u>	<u>Layer</u>		
$0.44n^2 + O(1)$	$0.66n$	(NT)	[Dolev et al. 1984]
$0.44n^2 + O(1)$	$0.66n$	(P3T)	[Fрати & Patrignani 2008]
$0.44n^2 + O(1)$	$0.66n$	(polyline)	[Bonichon et al. 2003]

- Straight-Line Drawing: Tight upper and lower bounds for layers (Area bounds can still be improved)
- Polyline Drawing: Tight upper and lower bounds both for area and layers
- Can we find better upper bounds in variable embedding setting?

Known Results

Fixed Embedding

<u>Upper Bounds</u>				<u>Lower Bounds</u>			
<u>Area</u>	<u>Layer</u>			<u>Area</u>	<u>Layer</u>		
$2n^2 + O(1)$	$n - 2$	(T)	[de Fraysseix et al. 1990]				
$n^2 + O(1)$	$n - 2$	(T)	[Schnyder 1990]				
$1.78n^2 + O(1)$	$0.66n$	(T)	[Chrobak and Nakano 1998]	$0.44n^2 + O(1)$	$0.66n$	(NT)	[Dolev et al. 1984]
$0.88n^2 + O(1)$	$0.66n$	(T)	[Brandenburg 2008]	$0.44n^2 + O(1)$	$0.66n$	(P3T)	[Fрати & Patrignani 2008]
$0.44n^2 + O(1)$	$0.66n$	(T)	(polyline) [Bonichon et al. 2003]	$0.44n^2 + O(1)$	$0.66n$	(polyline)	[Bonichon et al. 2003]

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Known Results

Fixed Embedding

Upper Bounds

Area

Layer

$2n^2 + O(1)$ $n - 2$ (T) [de Fraysseix et al. 1990]

$n^2 + O(1)$ $n - 2$ (T) [Schnyder 1990]

$1.78n^2 + O(1)$ $0.66n$ (T) [Chrobak and Nakano 1998]

$0.88n^2 + O(1)$ $0.66n$ (T) [Brandenburg 2008]

$0.44n^2 + O(1)$ $0.66n$ (T) (polyline) [Bonichon et al. 2003]

Lower Bounds

Area

Layer

$0.44n^2 + O(1)$ $0.66n$ (NT) [Dolev et al. 1984]

$0.44n^2 + O(1)$ $0.66n$ (P3T) [Frati & Patrignani 2008]

$0.44n^2 + O(1)$ $0.66n$ (polyline) [Bonichon et al. 2003]

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- Polyline Drawing: Tight upper and lower bounds both for area and layers
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Variable Embedding

Area

Layer

X $0.44n + O(1)$ (P3T) [Durocher & M. 2014]

$0.22n^2 + O(1)$ $0.33n$ (NT) [Frati and Patrignani 2008]

Known Results

Fixed Embedding

Upper Bounds

Area

Layer

$2n^2 + O(1)$ $n - 2$ (T) [de Fraysseix et al. 1990]

$n^2 + O(1)$ $n - 2$ (T) [Schnyder 1990]

$1.78n^2 + O(1)$ $0.66n$ (T) [Chrobak and Nakano 1998]

$0.88n^2 + O(1)$ $0.66n$ (T) [Brandenburg 2008]

$0.44n^2 + O(1)$ $0.66n$ (T) (polyline) [Bonichon et al. 2003]

Lower Bounds

Area

Layer

$0.44n^2 + O(1)$ $0.66n$ (NT) [Dolev et al. 1984]

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Variable Embedding

Area

Layer

X $0.44n + O(1)$ (P3T) [Durocher & M. 2014]

$0.22n^2 + O(1)$ $0.33n$ (NT) [Frati and Patrignani 2008]

$0.88n^2 + O(1)$ $0.66n$ (T) [Brandenburg 2008]

$0.44n^2 + O(1)$ $0.66n$ (T) (polyline) [Bonichon et al. 2003]

X $0.44n + O(\lambda\Delta)$ (T) (polyline) [Durocher & M. 2014]

Known Results

Fixed Embedding

Upper Bounds

Area	Layer		
$2n^2 + O(1)$	$n - 2$	(T)	[de Fraysseix et al. 1990]
$n^2 + O(1)$	$n - 2$	(T)	[Schnyder 1990]
$1.78n^2 + O(1)$	$0.66n$	(T)	[Chrobak and Nakano 1998]
$0.88n^2 + O(1)$	$0.66n$	(T)	[Brandenburg 2008]
$0.44n^2 + O(1)$	$0.66n$	(T) (polyline)	[Bonichon et al. 2003]

Lower Bounds

Area	Layer		
$0.44n^2 + O(1)$	$0.66n$	(NT)	[Dolev et al. 1984]
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- Can we find better upper bounds in variable embedding setting?

Variable Embedding

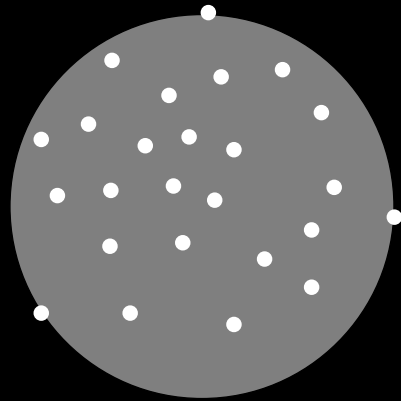
Area

Layer

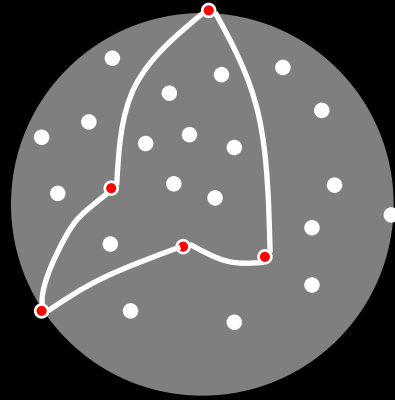
X	$0.44n + O(1)$	(P3T)	[Durocher & M. 2014]
$0.22n^2 + O(1)$	$0.33n$	(NT)	[Fрати and Patrignani 2008]
$0.88n^2 + O(1)$	$0.66n$	(T)	[Brandenburg 2008]
$0.44n^2 + O(1)$	$0.66n$	(T) (polyline)	[Bonichon et al. 2003]
X	$0.44n + O(\lambda\Delta)$	(T) (polyline)	[Durocher & M. 2014]

How to leverage variable embedding to compute a **layered polyline drawing** on smaller than $2n/3 \approx 0.66n$ layers?

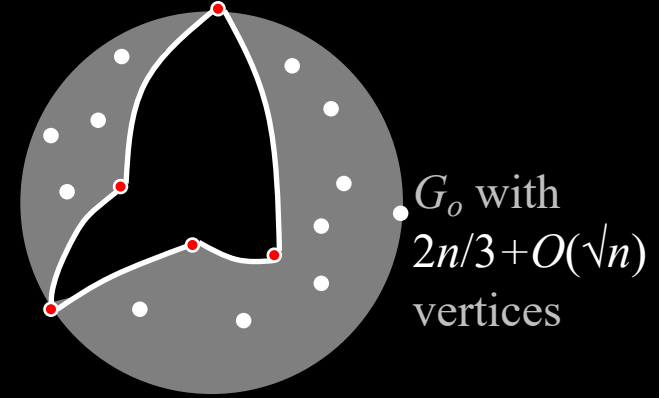
Previous Approach: $0.44n + O(\lambda\Delta)$ layers [Durocher & M. 2014]



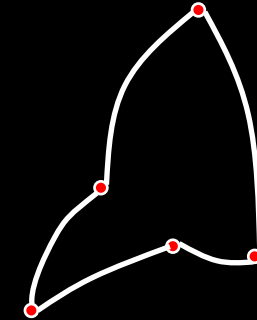
An n -vertex
planar graph G



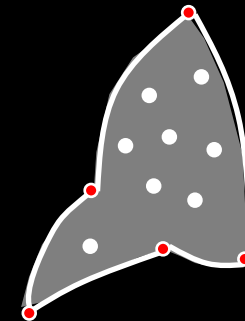
A simple cycle
separator of G



G_o with
 $2n/3 + O(\sqrt{n})$
vertices



A separator
of size $O(\sqrt{n})$

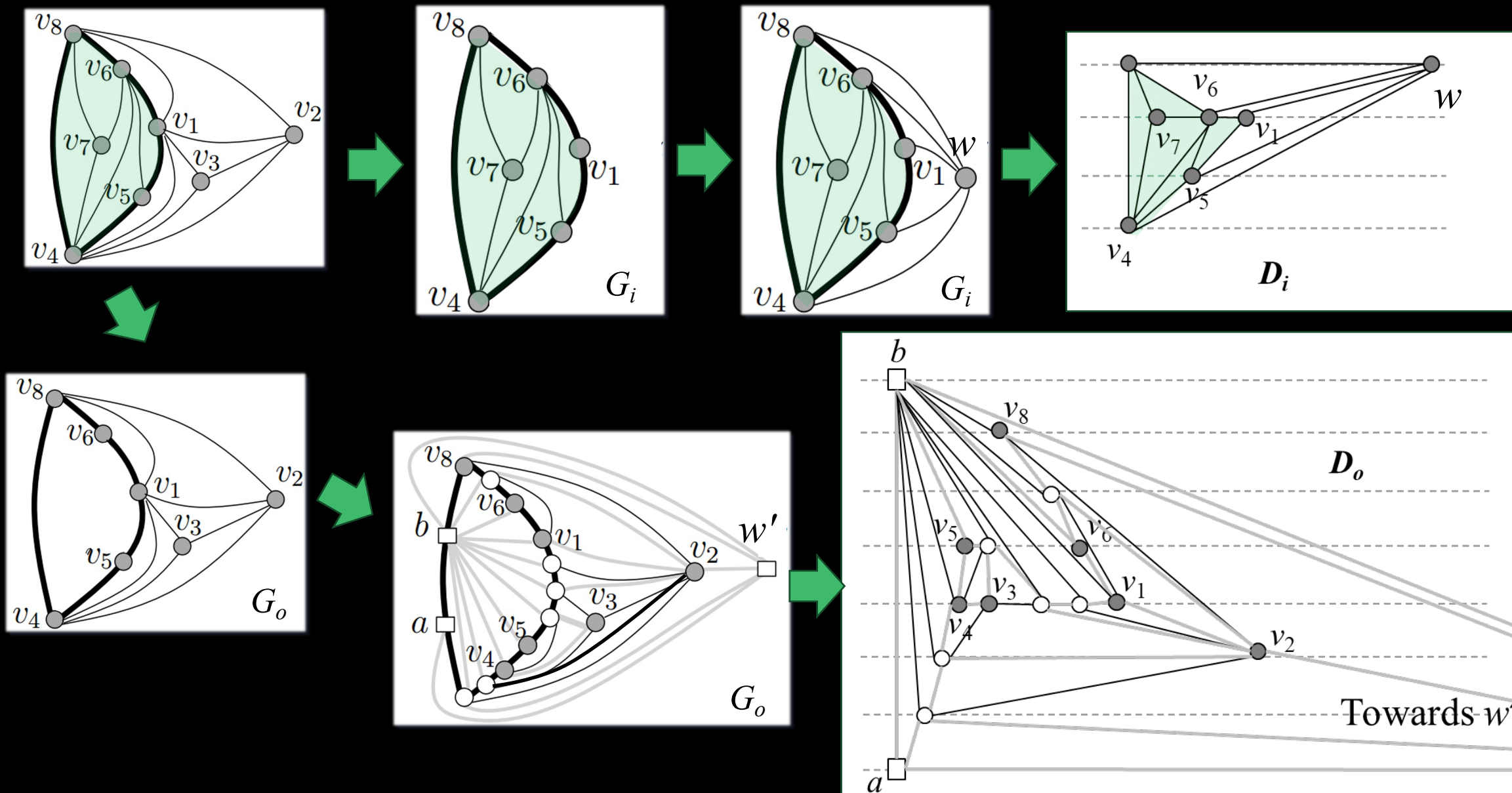


G_i with
 $2n/3 + O(\sqrt{n})$
vertices

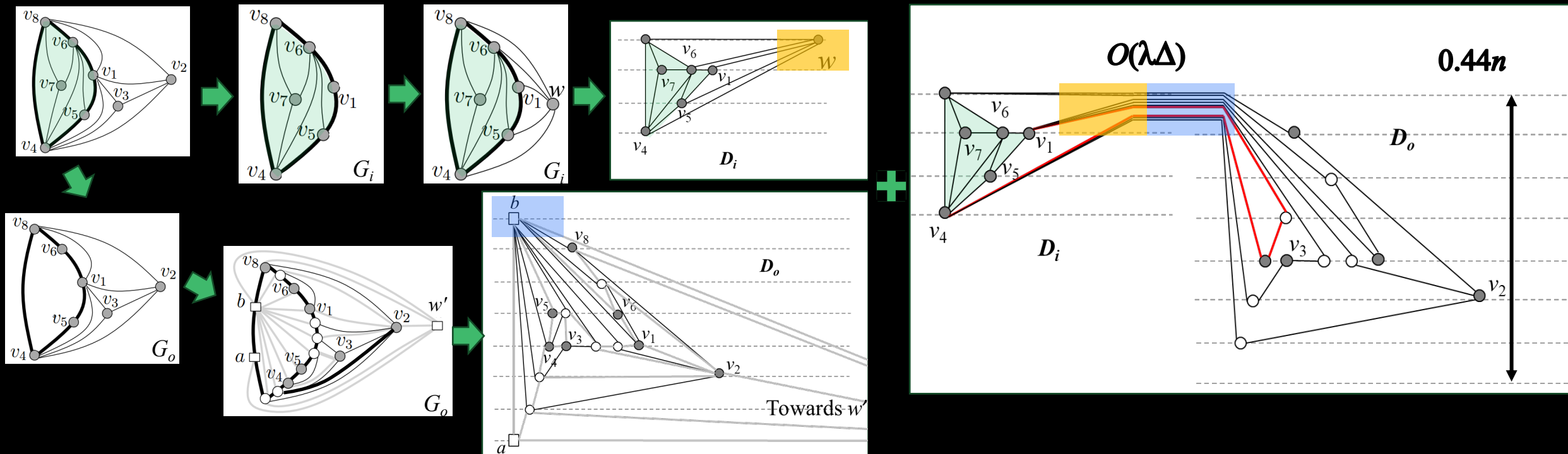
[Djidjev and Venkatesan, 1997]

Every planar triangulation has a simple cycle separator of size $O(\sqrt{n})$

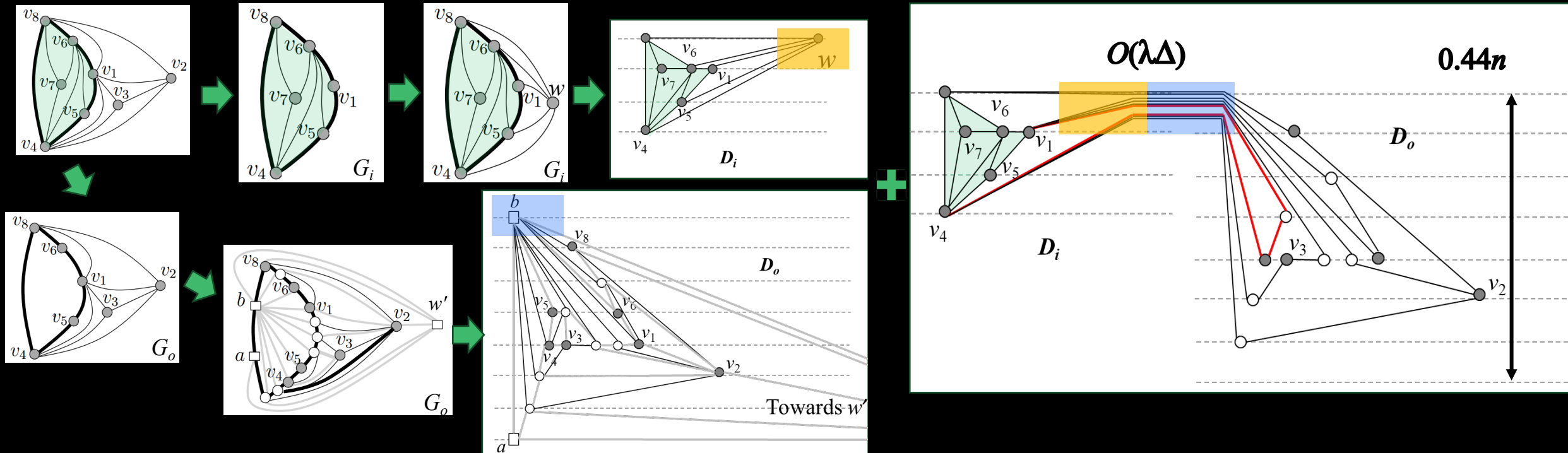
Previous Approach: $0.44n + O(\lambda\Delta)$ layers [Durocher & M. 2014]



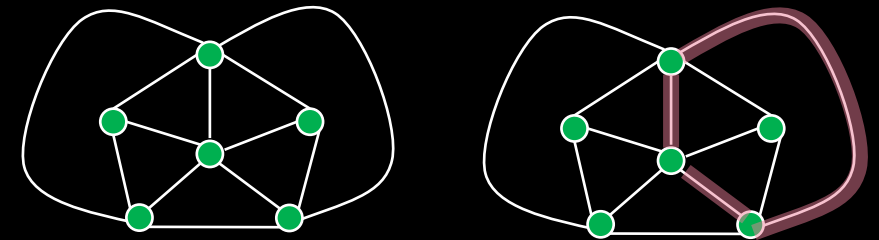
Previous Approach: $0.44n + O(\lambda\Delta)$ layers [Durocher & M. 2014]



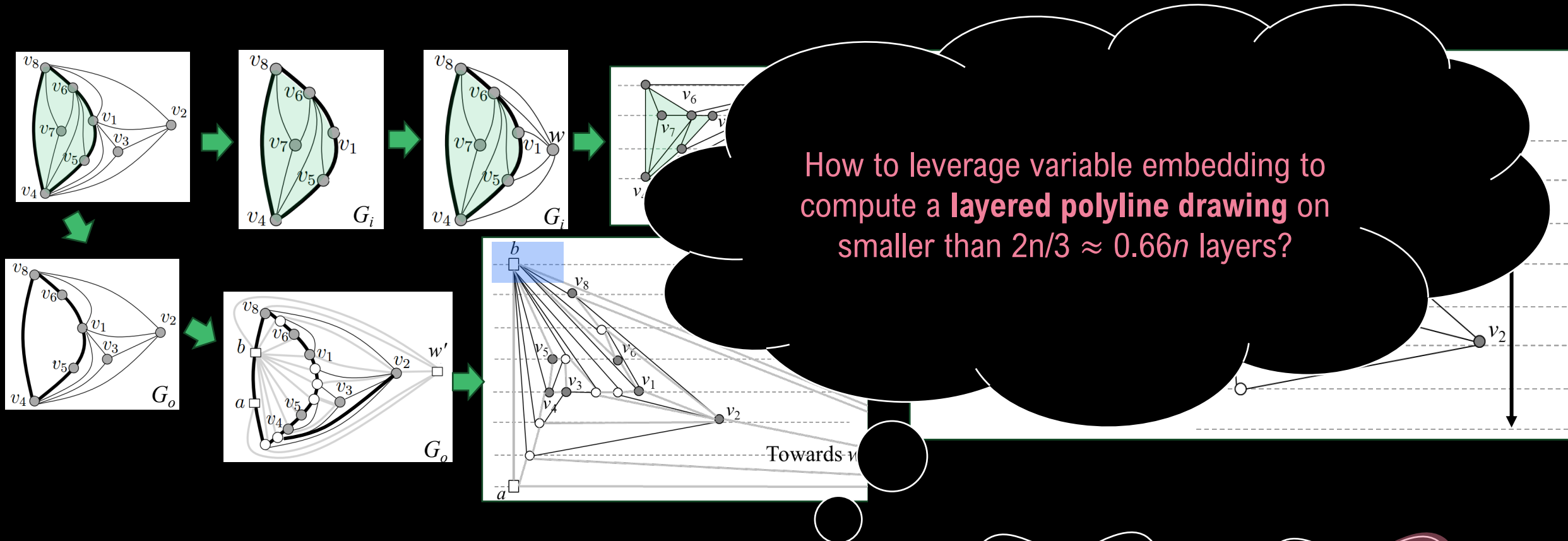
Previous Approach: $0.44n + O(\lambda\Delta)$ layers [Durocher & M. 2014]



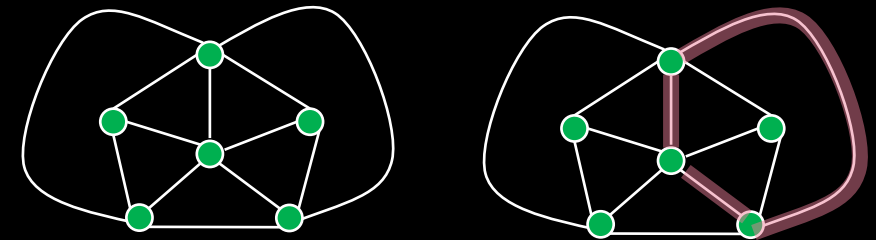
The term $O(\lambda\Delta)$ can sometimes be $\Omega(n)$. Consider the triangulation of a wheel graph; every cycle separator contains the center vertex.



Previous Approach: $0.44n + O(\lambda\Delta)$ layers [Durocher & M. 2014]

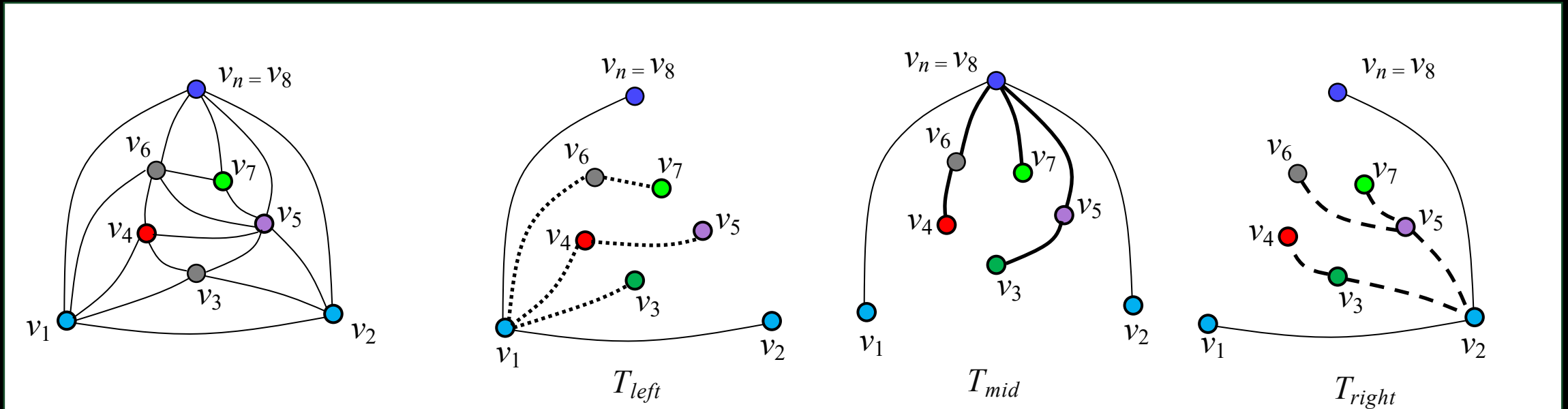


The term $O(\lambda\Delta)$ can sometimes be $\Omega(n)$. Consider the triangulation of a wheel graph; every cycle separator contains the center vertex.



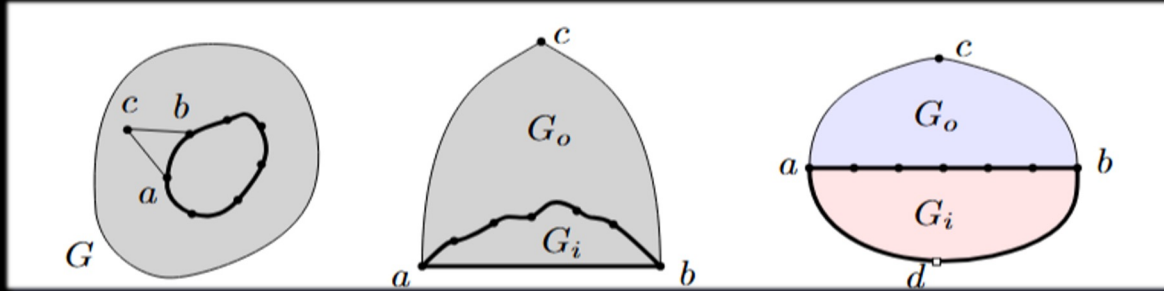
A Drawing on $14n/27 \approx 0.52n$ layers

Use cycle separator, draw and merge based on the trees of the minimum Schnyder realizer



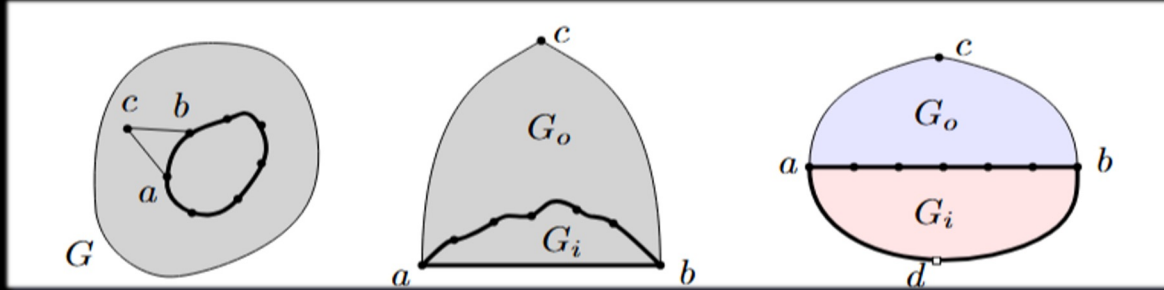
A Drawing on $14n/27 \approx 0.52n$ layers

Use cycle separator, draw and merge based on the trees of the minimum Schnyder realizer

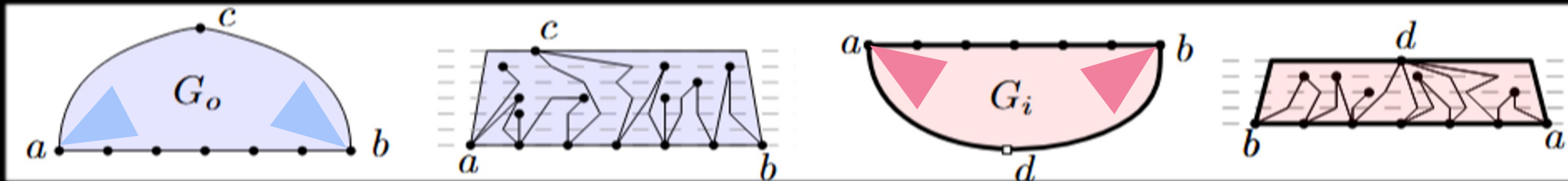


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Use cycle separator, draw and merge based on the trees of the minimum Schnyder realizer

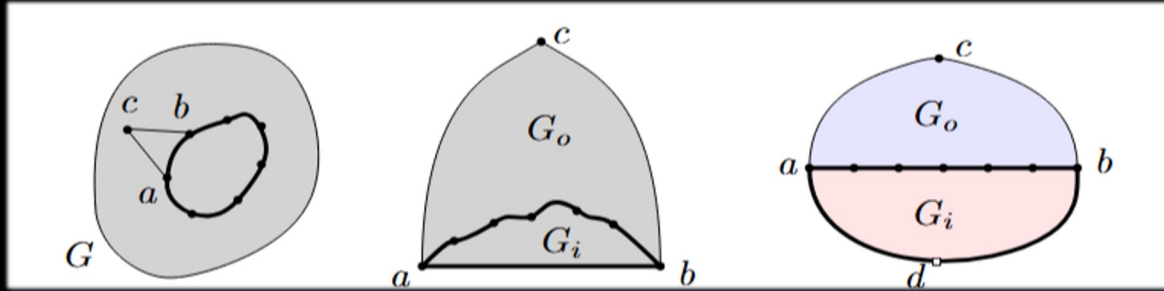


Case 1: T_a or T_b has small number of leaves ($< \alpha n$) in both G_o and G_i
 - Draw with a height of αn

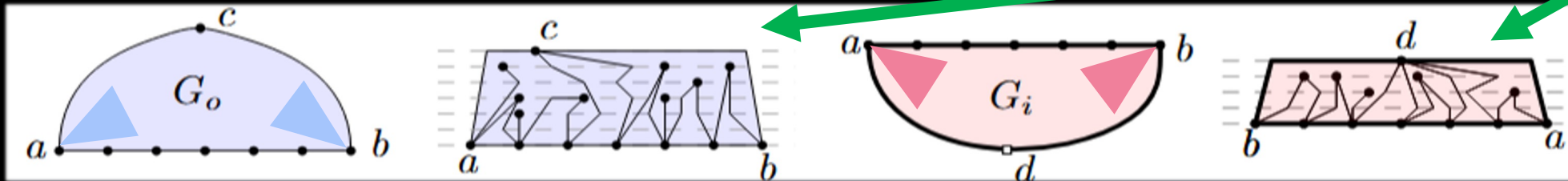
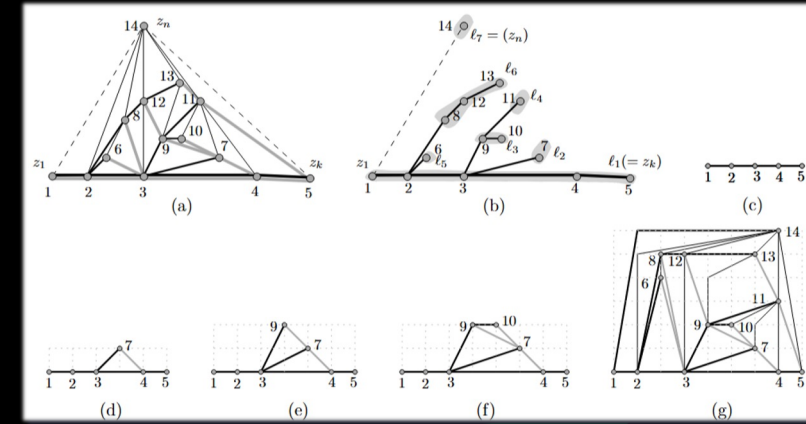


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Use cycle separator, draw and merge based on Schnyder realizer

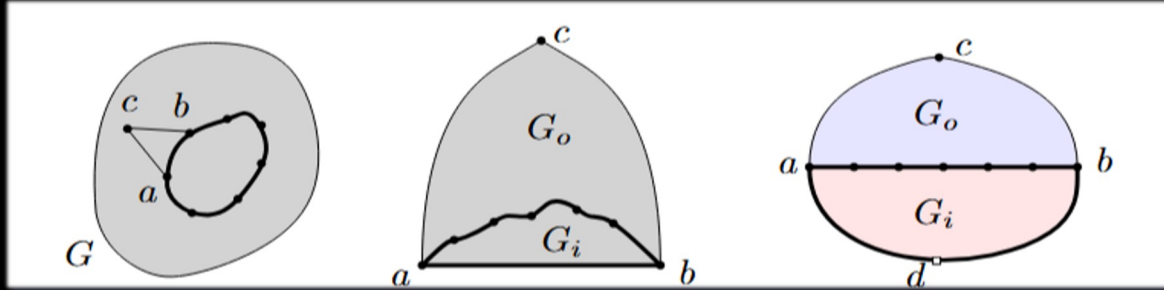


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 - Draw with a height of αn

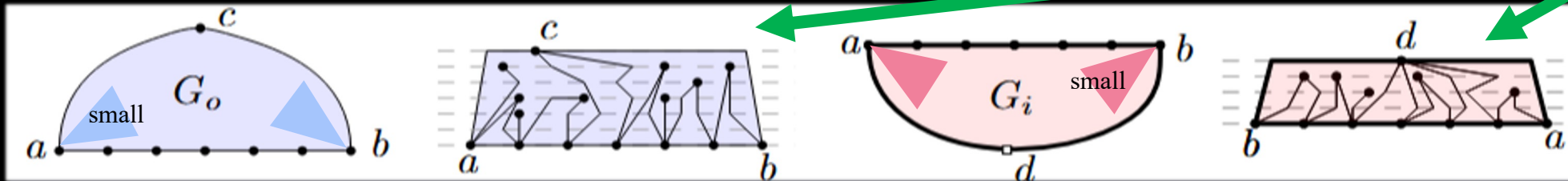
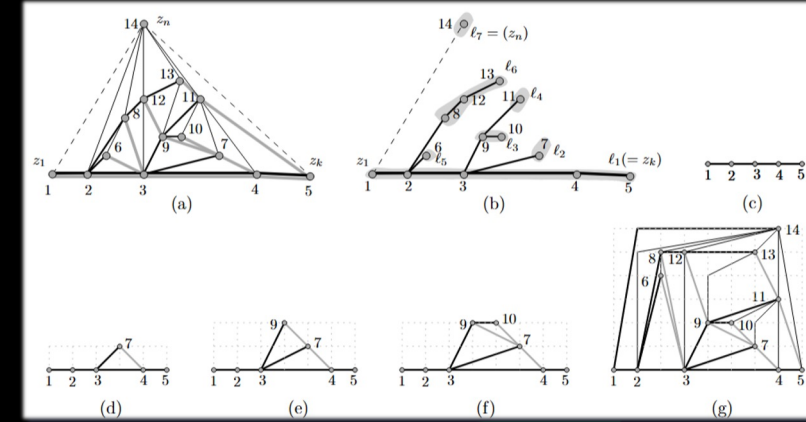


A Drawing on $14n/27 \approx 0.52n$ layers

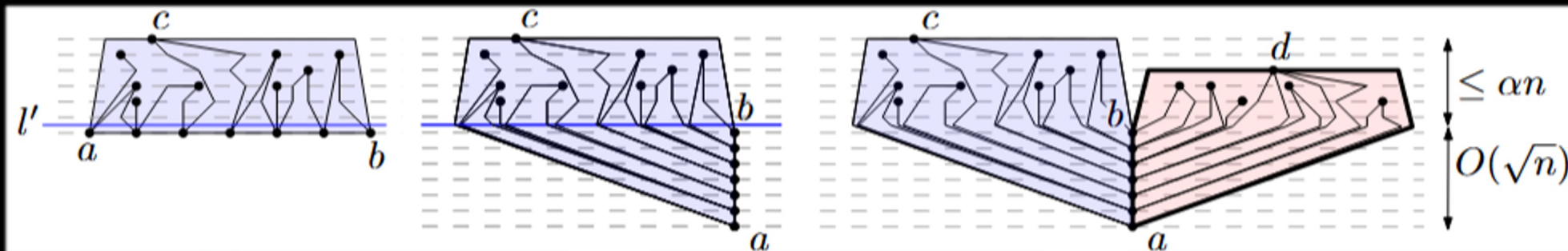
Use cycle separator, draw and merge based on Schnyder realizer



Case 1: T_a or T_b has small number of leaves ($< \alpha n$) in both G_o and G_i ,
- Draw with a height of αn

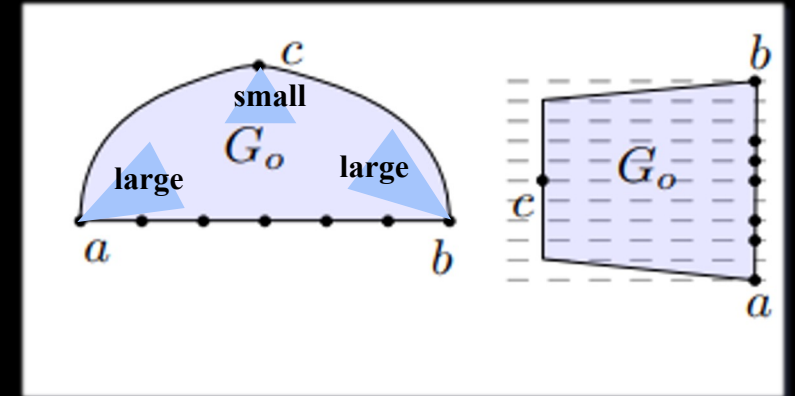


- Merge by pulling the $a, \dots, b$ path on a common vertical line

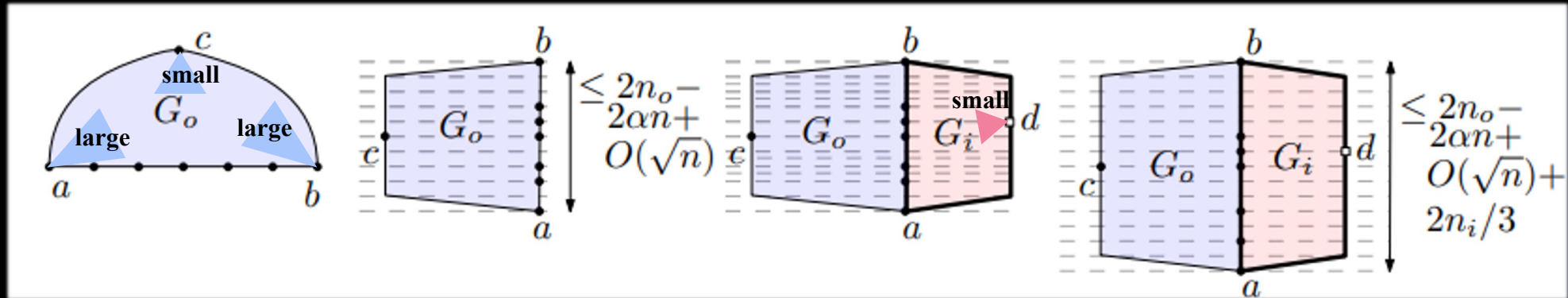


A Drawing on $14n/27 \approx 0.52n$ layers

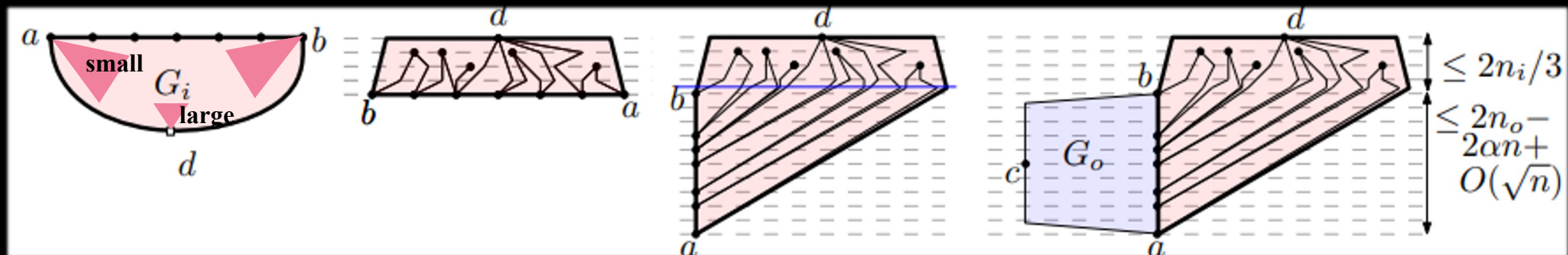
Case 2: T_a, T_b have more than αn leaves in at least one of G_o, G_i



Case 2.1: T_d has small number of leaves in G_i

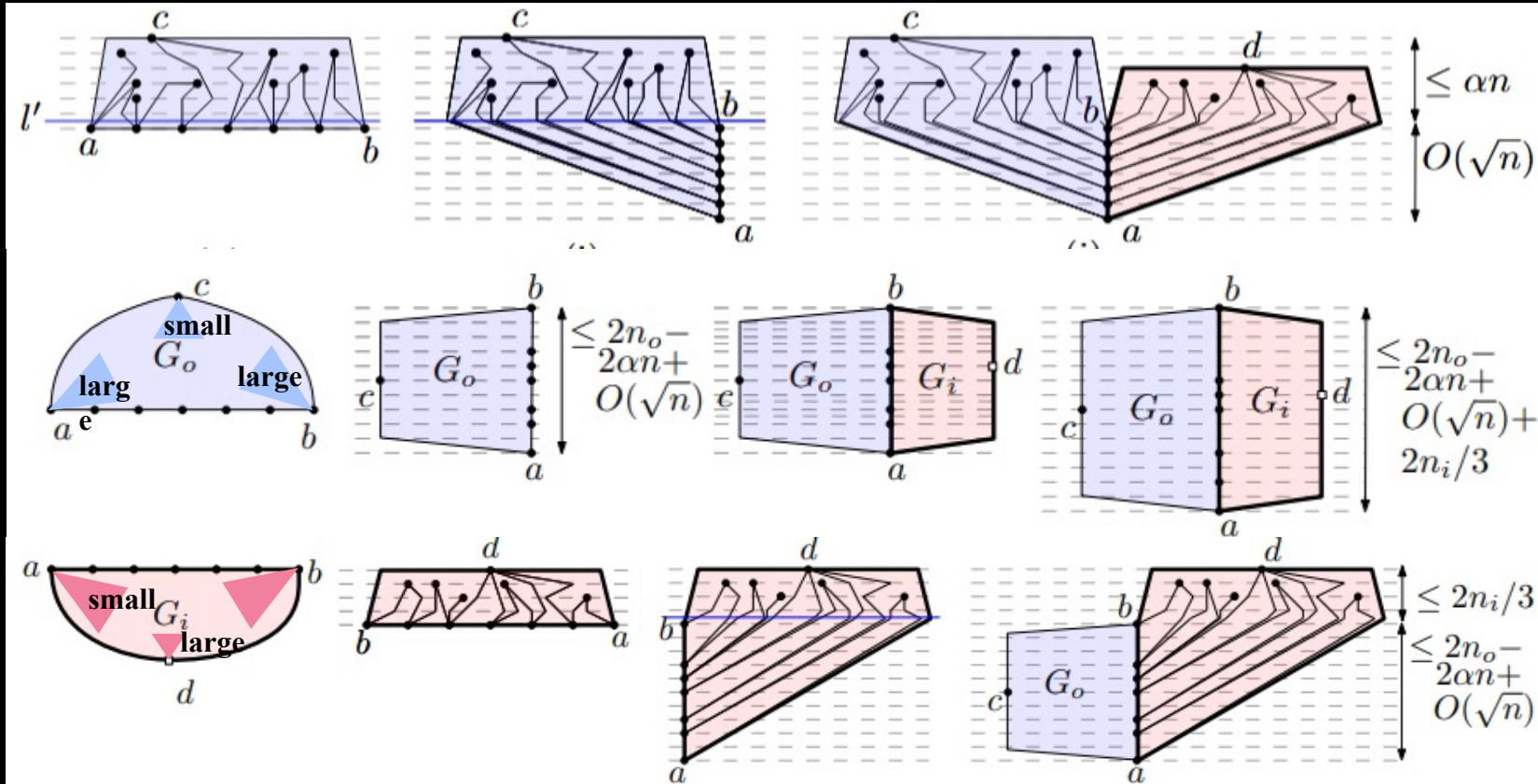


Case 2.2: T_d has large number of leaves in G_i



A Drawing on $14n/27 \approx 0.52n$ layers

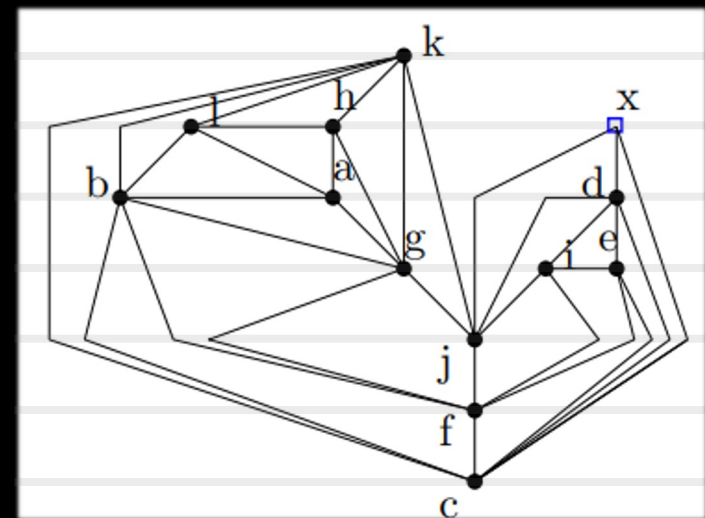
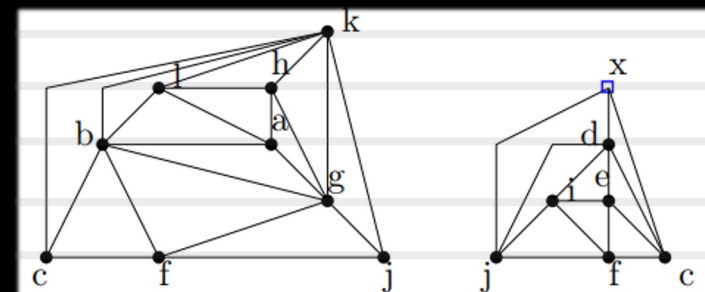
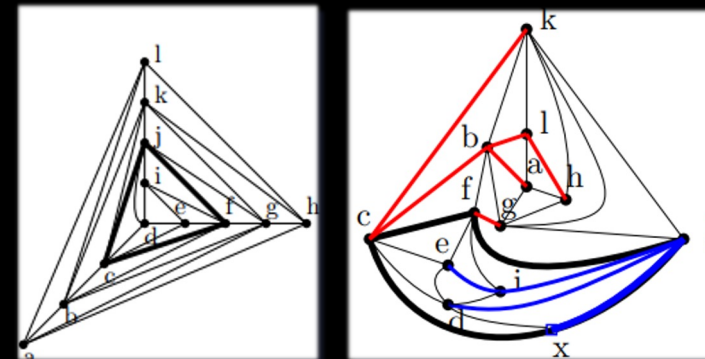
Choosing $\alpha = 14/27$ allows us to get the minimum over all cases



Every planar graph with n vertices admits a planar polyline drawing on $14n/27 + O(\sqrt{n})$ layers.

Summary and Open Questions

- Can we improve the bound? We are **still far from $n/3$ lower bound** in the variable embedding setting.
- We did not attempt to **reduce the number of bends** (currently, no more than **5 bends per edge**). It should be easy to bring it down to 3 but less than that is challenging with our technique.
- Can we get the **same $14n/27 + O(\sqrt{n})$ bound but with straight line edges?**
- Can we find polyline drawing in less than **$2n/3 \times 2n/3$ grid?**



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Thank You!

